



Enderass National PLC Digital Auction Management System

Software Requirements Specification (SRS)

Version: 2.0

1. Introduction

1.1 Purpose

The purpose of this system is to digitize and manage the core auction lifecycle for Enderass National PLC through a web application and mobile applications. The Platform Release focuses on essential auction operations, including user registration, asset submission, evaluation, auction publication, document sales, payment processing, CPO management, bid submission, winner selection, notifications, reporting, and administration.

The system shall support:

- Web Application
- Android Mobile Application
- iOS Mobile Application

Supported Languages:

- English
- Amharic

1.2 Objectives

The Platform Release aims to:

- Reduce manual auction processes.
- Improve transparency and accountability.
- Enable online bidder participation.
- Digitize asset evaluation and auction management.
- Provide secure bid submission and winner selection.
- Support online and manually verified payments.
- Provide operational reporting and dashboards.

1.3 Intended Audience

This document is intended for:

- Business Stakeholders
- Project Sponsors
- Product Owners
- Business Analysts
- Software Developers
- QA/Test Engineers
- System Administrators

2. Scope

The Platform Release shall include the following modules:

1. User Registration & Authentication
2. Basic KYC Verification
3. Asset Owner Management
4. Asset Submission & Ownership Verification
5. Asset Evaluation Management
6. Auction Management
7. Auction Document Management
8. Payment Management
9. Simplified CPO Management
10. Closed Bid Submission
11. Winner Selection
12. Notification Management
13. Reporting & Dashboard
14. Staff Management
15. Role-Based Access Control
16. Android Mobile Application
17. iOS Mobile Application

3. Stakeholders

3.1 External Users

- Individual Bidders
- Asset Owners
- Private Organizations
- Banks
- Insurance Companies
- NGOs
- Government Institutions

3.2 Internal Users

- Super Administrator
- Auction Manager
- Evaluation Officer
- Finance Officer
- Customer Service Officer

4. Business Workflow

Platform: User Registration & KYC

1. User registers an account.
2. User verifies mobile number using OTP.
3. User uploads identification documents.
4. Staff reviews KYC documents.
5. User account is approved or rejected.

Phase 2: Asset Submission

1. Asset owner submits asset details.
2. Ownership documents are uploaded.
3. Staff reviews ownership documents.
4. Asset is approved for evaluation.

Phase 3: Asset Evaluation

1. Evaluation officer schedules inspection.
2. Asset valuation is recorded.
3. Photos and evaluation report are uploaded.
4. Evaluation is approved or rejected.

Phase 4: Auction Creation

1. Auction manager creates auction.
2. Approved asset is linked to auction.
3. Auction documents are uploaded.
4. Auction is published.

Phase 5: Participation

1. Bidder purchases auction documents.
2. Payment is completed through Addis Pay or manual payment.
3. Payment is verified.
4. Bidder uploads CPO document.
5. CPO is reviewed and approved.
6. Bidder submits bid.

Phase 6: Auction Closure

1. Auction closes automatically at the configured date and time.
2. Bid submission is disabled.
3. Submitted bids become read-only.

Phase 7: Winner Selection

1. Authorized staff review submitted bids.

2. Highest valid bid is identified.
3. Winner is approved.
4. Notifications are sent.

5. Functional Requirements

5.1 User Registration & Authentication

FR-AUTH-001

Users shall register using mobile number verification through OTP.

FR-AUTH-002

The system shall prevent duplicate accounts using:

- Mobile Number
- National ID Number
- TIN Number (where applicable)

FR-AUTH-003

Organization users shall upload:

- Trade License
- TIN Certificate
- Business Registration Certificate

FR-AUTH-004

Users shall log in using mobile number/email and password.

FR-AUTH-005

Users shall be able to reset passwords using OTP verification.

FR-AUTH-006

Passwords shall be stored using secure password hashing.

5.2 Basic KYC Verification

FR-KYC-001

Users shall upload identification documents during registration.

FR-KYC-002

Staff shall review submitted KYC documents.

FR-KYC-003

Staff shall approve or reject KYC submissions.

FR-KYC-004

Users shall receive notifications regarding KYC status changes.

5.3 Asset Owner Management

FR-OWNER-001

Asset owners shall maintain profile information.

FR-OWNER-002

Asset owners shall view submitted assets.

FR-OWNER-003

Asset owners shall track asset approval status.

5.4 Asset Submission & Ownership Verification

FR-ASSET-001

Asset owners shall submit asset information.

FR-ASSET-002

Asset owners shall upload ownership documents.

Supported ownership documents include:

Vehicles:

- Vehicle Registration Book

Land:

- Title Deed

Buildings:

- Ownership Certificate

Machinery:

- Purchase Documents

FR-ASSET-003

Staff shall review ownership documents.

FR-ASSET-004

Assets shall be approved or rejected before evaluation.

FR-ASSET-005

Users shall receive notifications regarding asset review decisions.

5.5 Asset Evaluation Management

FR-EVAL-001

Evaluation officers shall schedule inspections.

FR-EVAL-002

Evaluation officers shall record valuation details.

FR-EVAL-003

Evaluation officers shall upload photographs.

FR-EVAL-004

Evaluation officers shall upload evaluation reports.

FR-EVAL-005

Evaluation officers shall recommend approval or rejection.

FR-EVAL-006

Only approved evaluations may proceed to auction creation.

5.6 Auction Management

FR-AUC-001

Auction managers shall create auctions.

FR-AUC-002

Auctions shall be linked to approved assets.

FR-AUC-003

Auction managers shall define:

- Auction Title
- Auction Description
- Start Date
- End Date

- Reserve Price
- Document Price

FR-AUC-004

Auctions shall be published after approval.

FR-AUC-005

Published auctions shall be visible to bidders.

FR-AUC-006

Suspended auctions shall prevent further participation.

5.7 Auction Document Management

FR-DOC-001

Staff shall upload auction documents.

FR-DOC-002

Documents shall be downloadable only after payment verification.

FR-DOC-003

The system shall track document downloads.

FR-DOC-004

Supported document formats shall include PDF.

5.8 Payment Management

FR-PAY-001

The system shall integrate with Addis Pay as the primary online payment gateway.

FR-PAY-002

Users shall be able to upload payment receipts for manual payment verification.

FR-PAY-003

Finance officers shall review uploaded payment receipts.

FR-PAY-004

Finance officers shall approve or reject payments.

FR-PAY-005

The system shall maintain payment statuses:

- Pending
- Approved
- Rejected

FR-PAY-006

Users shall receive payment confirmation notifications.

FR-PAY-007

The system shall generate transaction reports.

FR-PAY-008

Verified payments shall grant access to auction documents.

5.9 Simplified CPO Management

FR-CPO-001

Users shall upload CPO documents.

FR-CPO-002

Authorized staff shall review submitted CPO documents.

FR-CPO-003

Authorized staff shall approve or reject CPO submissions.

FR-CPO-004

The system shall maintain CPO statuses:

- Pending
- Approved
- Rejected

FR-CPO-005

Users shall receive notifications regarding CPO decisions.

5.10 Closed Bid Submission

FR-BID-001

Approved bidders shall submit bids before auction closing time.

FR-BID-002

Users shall only view their own submitted bids.

FR-BID-003

Submitted bids shall not be editable.

FR-BID-004

The system shall automatically close bid submission when the auction end time is reached.

FR-BID-005

The system shall record bid submission timestamps.

5.11 Winner Selection

FR-WIN-001

Authorized staff shall review submitted bids after auction closure.

FR-WIN-002

The system shall identify the highest valid bid.

FR-WIN-003

The selected winner shall be recorded.

FR-WIN-004

Winner notifications shall be sent automatically.

5.12 Notification Management

FR-NOT-001

The system shall send notifications for:

- Registration
- KYC Approval/Rejection
- Asset Approval/Rejection
- Payment Approval/Rejection
- CPO Approval/Rejection
- Auction Publication
- Winner Announcement

FR-NOT-002

Notifications shall be delivered through:

- SMS
- Email
- In-App Notifications

5.13 Reporting & Dashboard

FR-REP-001

The system shall provide dashboard summaries.

FR-REP-002

The system shall generate reports including:

- Active Auctions
- Closed Auctions
- Asset Reports
- Bidder Reports
- Payment Reports
- CPO Reports

FR-REP-003

Reports shall support:

- Date Filters
- Export to PDF
- Export to Excel

5.14 Staff Management

FR-STAFF-001

Administrators shall create staff accounts.

FR-STAFF-002

Administrators shall activate or deactivate staff accounts.

FR-STAFF-003

Administrators shall assign roles to staff members.

5.15 Role-Based Access Control

FR-RBAC-001

The system shall support role-based permissions.

FR-RBAC-002

Users shall only access authorized functions.

FR-RBAC-003

Roles shall include:

- Super Administrator
- Auction Manager
- Evaluation Officer
- Finance Officer
- Customer Service Officer
- Bidder
- Asset Owner

6. Business Rules

BR-001

Only approved users may participate in auctions.

BR-002

Only approved assets may be auctioned.

BR-003

Only verified payments shall grant document access.

BR-004

Only approved CPO submissions shall allow bid submission.

BR-005

Bids submitted after auction closure shall be rejected.

BR-006

Highest valid bid wins.

BR-007

Reserve price must be met.

BR-008

Tie bids shall be resolved using the earliest submission timestamp.

BR-009

If the winner declines, the next highest valid bidder may be selected.

7. Data Model and ERD Considerations

The detailed ERD shall be produced during system design.

Core entities include:

1. User
2. Role
3. KYC Verification
4. Asset Owner
5. Asset
6. Ownership Verification
7. Evaluation
8. Auction
9. Auction Document
10. Payment
11. CPO
12. Bid
13. Winner
14. Notification
15. Staff
16. Audit Log

Key Relationships:

- One User may own multiple Assets.
- One Asset shall have one Evaluation.
- One Asset may belong to one Auction.
- One Auction may have multiple Bids.
- One Bidder may submit one Bid per Auction.
- One Auction may have one Winner.
- One User may have multiple Payments.
- One User may have multiple Notifications.

8. Integration Requirements

8.1 Payment Gateway

Addis Pay Integration

The system shall:

- Initiate payments through Addis Pay.
- Receive payment confirmation responses.
- Record transaction references.
- Update payment status.

8.2 Communication Services

The system shall integrate with:

- SMS Gateway
- Email Service

8.3 File Storage

The system shall support secure storage of:

- KYC Documents
- Ownership Documents
- Evaluation Reports
- Auction Documents
- Payment Receipts
- CPO Documents

9. Security Requirements

SEC-001

All communication shall use HTTPS.

SEC-002

Passwords shall be securely hashed.

SEC-003

Role-based authorization shall be enforced.

SEC-004

Failed login attempts shall be logged.

SEC-005

User activities shall be recorded in audit logs.

SEC-006

Uploaded files shall be validated for type and size.

SEC-007

Session timeout shall be enforced after inactivity.

10. Mobile Application Requirements

Android Application

Supported Version:

- Android 10 and above

iOS Application

Supported Version:

- iOS 15 and above

Mobile Features

- User Registration
- Login
- Profile Management
- Asset Submission
- Document Upload
- Auction Browsing
- Auction Document Purchase
- Payment Tracking
- CPO Upload
- Bid Submission
- Notifications
- Dashboard Access

11. Non-Functional Requirements

Performance

- Page load time shall be less than 5 seconds under normal load.
- Bid submission shall complete within 5 seconds.

- Payment status updates shall be reflected within 1 minute after confirmation.

Availability

- System availability shall be 99% uptime.

Backup

- Daily database backup.
- Daily document backup.

Scalability

- Architecture shall support future expansion without major redesign.

12. User Stories

US-001

As a bidder, I want to register and verify my account so that I can participate in auctions.

US-002

As an asset owner, I want to submit assets and ownership documents so that they can be auctioned.

US-003

As an evaluation officer, I want to record asset valuations so that auctions can be created.

US-004

As a bidder, I want to purchase auction documents so that I can participate in bidding.

US-005

As a finance officer, I want to verify payments so that only eligible bidders can access auction documents.

US-006

As a bidder, I want to upload my CPO document so that I can submit a bid.

US-007

As a bidder, I want to submit a bid securely before auction closure.

US-008

As an auction manager, I want to determine the winner after auction closure.

13. Acceptance Criteria

Submit Asset

1. Asset owner can submit asset details.
2. Ownership documents can be uploaded.
3. Asset status is recorded as Pending Review.
4. Audit log is generated.

Verify Payment

1. Finance officer can review payment receipt.
2. Payment can be approved or rejected.
3. User receives notification.
4. Payment status is updated.

Submit Bid

1. Approved bidder can submit a bid.
2. Bid cannot be modified after submission.
3. Bid timestamp is recorded.
4. Audit log is generated.

Select Winner

1. System identifies highest valid bid.
2. Winner is recorded.
3. Winner notification is sent.
4. Auction status changes to Closed.

14. Future Enhancements

The following features are planned for future releases:

- Automated CPO Calculation
- Automated CPO Expiration Monitoring
- Advanced Analytics
- AI Asset Valuation
- Digital Signature
- Live Auctions
- Open Bidding
- National Auction Marketplace
- Government Asset Auction Portal
- Multi-Tenant Architecture
- Multi-Company Support

- Multi-Branch Support

15. Success Metrics

- Reduction in manual paperwork.
- Faster auction processing time.
- Increased bidder participation.
- Improved transparency.
- Improved payment tracking.
- Complete auditability of auction activities.
- Successful adoption of web and mobile applications.